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History of Astronomy: The Exoplanet Revolution

New Worlds Beyond

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Introduction

For all of recorded history, the answer to the oldest question in astronomy, are there other worlds, was a matter of faith and argument alone. The philosophers argued it: the atomists of Greece posited infinite worlds, Aristotle denied them, Bruno died for asserting them, and the astronomers of the telescope era, finding the planets and the stars, still found no planet but the ones circling the Sun. By the close of the last century the argument had become a standing embarrassment: the physics said planets should be common, the nebular theory of the Sun's formation said the same, and the observational record held one example, our own, a sample of one from which nothing could be concluded. The question was not whether the other worlds existed but whether the human species would ever be clever enough to see one.

Then, in a single decade, the wall fell. The first confirmed planets, found in 1992 around a pulsar, the collapsed remnant of a supernova, announced that the search had begun in earnest; the first planet around a star like the Sun, found in 1995 around 51 Pegasi, revealed a world no theory had predicted, a Jupiter orbiting closer to its star than Mercury to ours, and the surprise opened the field's floodgates; and the great surveys that followed, the radial-velocity hunters and the space telescopes, brought the count from zero to thousands in a single generation. The exoplanet revolution, this book's subject, is one of the swiftest and most complete transformations in the history of science: a question open for two and a half thousand years, answered in twenty.

This book follows the revolution in full. It begins with the search's long failure and the physics that made the failure a puzzle, the wobble methods and the early false alarms, and the two discoveries that broke the dam; it follows the techniques, the radial velocity, the transit, the microlens, the direct image, the machinery by which a planet's gravity or its shadow or its light is read from a star's; it surveys the strange harvest, hot Jupiters and super-Earths and mini-Neptunes, lava worlds and water worlds, systems packed tighter than anyone dared predict and worlds in the temperate zone where life might sit; and it follows the science that the harvest built, the migration of planets, the demographics of the planetary zoo, the atmospheric spectroscopy that is now reading the skies of other worlds, and the long push toward the question the revolution has always served: the detection of a planet like Earth, and the search for signs of life upon it.

A word on the book's coverage and its figures, before the voyage of the chapters begins. The account draws on the field's standard sources: the discovery papers and the mission archives, the demographic syntheses and the theoretical reviews, and the community's catalogs, the public tallies of the confirmed worlds that the field maintains as its census. The figures quoted are the science's own settled results, the occurrence rates and the census milestones, stated as the literature states them, rounded where the field rounds; the named worlds, the classes' representatives, are the census's documented members; and where the science is open, the habitability debates and the biosignature's standards, the account says so plainly. The revolution's own vocabulary, the radial velocity and the transit, the super-Earth and the habitable zone, is the field's own coinage, and the chapters that follow use it as the field does, defining each term at its first use, so that the reader finishes the book holding the language of the subject as well as its story.

The revolution's meaning, to be held from the first page, is double: it is a scientific event, the opening of an entire field's subject matter, and it is a human one, the answer to the oldest question, and the answer's form, a cosmos of uncounted worlds, has already changed the way the species thinks about itself. The chapters that follow tell the story from the failures to the frontier, and the reader who finishes them will know both the science of the exoplanets and the shape of the age that found them.

Chapter 1: The Discovery of The Exoplanet Revolution

The discovery has a long preface, and it begins with the imagining. The idea of other worlds is older than the telescope: the Greek atomists, Democritus and Epicurus and their school, argued for infinite worlds in an infinite universe, while Aristotle's cosmos held the Earth unique at the center; the medieval age inherited both positions, and the question survived, in theology and poetry, to the beginning of the modern era. The Copernican revolution made the question sharper: if the Earth was a planet, the planets were worlds, and the stars, if they were suns, might have their own; the astronomers of the seventeenth and eighteenth centuries, Huygens and Kant and Herschel among them, argued the plurality of worlds as the natural reading of the new cosmos, and the nebular hypothesis of the Sun's formation, the theory that the solar system condensed from a rotating cloud, made planets a by-product of star formation itself. The reasoning, by the nineteenth century, was overwhelming: planets should be everywhere, and the only missing piece was the observation.

The observation's failure, the century of nothing, is the discovery's second preface, and it deserves honest telling. The methods existed in principle: a planet tugs its star, so the star should wobble; a planet blocks light, so the star should dim; the wobble and the dim, in the astronomers' calculations, were at the very edge of the instruments' reach, and the failure to find them was read, by the field's pessimists, as the universe's answer. The first claimed detections, in the mid-twentieth century, added to the confusion: the Barnard's Star wobble, announced by the astrometrist Peter van de Kamp in the 1960s, was a celebrated candidate whose planets dissolved, over decades, into telescope error; other claims came and went, and the field, by the 1980s, was known for its alarms more than its discoveries.

The discovery proper arrived in two strokes, a few years apart. In 1992, the radio astronomers Alexander Wolszczan and Dale Frail, timing the pulses of a dead star, a millisecond pulsar in the constellation Virgo, found the pulses arriving early and late in a periodic rhythm, the signature of planets: two, then three, Earth-sized worlds orbiting the pulsar, the remnants of a system destroyed by the supernova, and the first planets ever confirmed around another star. The discovery, momentous as it was, left the main question open: pulsar planets, lit by the deadly beam of their dead sun, were nobody's image of a habitable world, and the field waited for the real thing.

A further thread of the discovery deserves the chapter's closing mention, the near misses, the detections that history almost had. The early wobble surveys of the 1980s, the Canadian and American programs, recorded the signals of real planets and doubted them: the Gamma Cephei wobble, measured in the decade's data and shelved as the star's own variation, was confirmed two decades later as a planet, the giant world the 1980s had found and declined; the timing residuals of other stars, the same story repeated, the planets hiding in the error bars of instruments not yet trusted. The thread's lesson is the discovery's own physics: the signals were there all along, at the edge of the machines' reach, and the revolution's timing was set by the instruments' crossing of the threshold, a few years of engineering between the near miss and the name in the journals. The reader should hold the thread as the history's humility: the universe was speaking, and the species, one generation earlier, was almost able to hear it.

A last layer of the discovery deserves its paragraph, the technology's prehistory, the decades of patient building that made 1995 possible. The echelle spectrographs of the era, the optical machines that spread starlight into its thousands of lines; the calibration research, the reference lamps and the stable cells, the iodine absorption whose fixed lines anchored the measurements; the detector revolution, the CCDs that replaced the photographic plate and its chemistry with the digital counts: the prehistory's through-line is precision, each decade's engineering buying the next decimal, and the discovery's historians, tracing the lineage, credit the machine as much as the observers, the spectrograph that measured the wobble being a half-century's accumulation of optics, electronics, and stubbornness. The layer's lesson is the discovery's own dependence: 51 Pegasi b was found because the instrument could find it, and the instrument could find it because a generation of engineers, uncelebrated, had built the machine that the moment required.

It came in 1995, from Switzerland. Michel Mayor and Didier Queloz, running a high-precision spectrograph on the Haute-Provence Observatory in southern France, had surveyed stars like the Sun for the wobble, and found one: 51 Pegasi, a sunlike star fifty light years away, oscillating in a four-day rhythm, the signature of a companion half the mass of Jupiter circling at a twentieth of Earth's own solar distance. The planet, 51 Pegasi b, was an impossibility by the theory of the day, a Jupiter in Mercury's orbit, and the discovery paper's reception, the checks and the confirmations that followed within weeks, ended with the astronomy community accepting the impossible and rewriting the theory: the giant planets moved, and the solar system was not the template the field had assumed. The two discoveries, the pulsar worlds and the hot Jupiter, opened the revolution's gate, and the decade that followed, the finding of dozens, then hundreds of worlds, made the gate's opening into the flood this book describes.